

TABLE 5.20 First Assignment in VAM ($x_{31} = 5$)

	1	2	3	4	Row penalty
1	10	15	20	0	10
2	12	7	15	10	10
3	5	4	14	16	5
Column penalty	5	1	5	15	

VAM

	15		0
		15	10
5			5

Feasible
Basic.
degenerate

$$m+n-1=6.$$

TABLE 5.21 Starting Iteration

	1	2	3	4	Supply
1	10	2	20	11	15
2	12	7	9	20	25
3	4	14	16	10	10
Demand	5	15	15	15	

$$2 = 520.$$

$$u_i + v_j = c_{ij}$$

Basic variable	(u, v) Equation	Solution
x_{11}	$u_1 + v_1 = 10$	Set $u_1 = 0 \rightarrow v_1 = 10$
x_{12}	$u_1 + v_2 = 2$	$u_1 = 0 \rightarrow v_2 = 2$
x_{22}	$u_2 + v_2 = 7$	$v_2 = 2 \rightarrow u_2 = 5$
x_{23}	$u_2 + v_3 = 9$	$u_2 = 5 \rightarrow v_3 = 4$
x_{24}	$u_2 + v_4 = 20$	$u_2 = 5 \rightarrow v_4 = 15$
x_{34}	$u_3 + v_4 = 18$	$v_4 = 15 \rightarrow u_3 = 3$

6 equations.
7 unknowns) 1 arbitrary value.

In TP alg, $u_1 = 0$.

Next, we use u_i and v_j to evaluate the nonbasic variables by computing

$$u_i + v_j - c_{ij}, \text{ for each nonbasic } x_{ij} \text{ empty cell.}$$

Nonbasic variable = x_{ij}	$u_i + v_j - c_{ij}$
x_{13}	$u_1 + v_3 - c_{13} = 0 + 4 - 20 = -16$
x_{14}	$u_1 + v_4 - c_{14} = 0 + 15 - 11 = 4$
x_{21}	$u_2 + v_1 - c_{21} = 5 + 10 - 12 = 3$
x_{31}	$u_3 + v_1 - c_{31} = 3 + 10 - 4 = 9$
x_{32}	$u_3 + v_2 - c_{32} = 3 + 2 - 14 = -9$
x_{33}	$u_3 + v_3 - c_{33} = 3 + 4 - 16 = -9$

The preceding information, together with the fact that $u_i + v_j - c_{ij} = 0$ for each basic x_{ij} , is actually equivalent to computing the **z-row of the simplex tableau**, as the following summary shows.

positive z-row values in min. E.V.

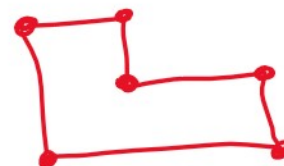
Basic	x_{11}	x_{12}	x_{13}	x_{14}	x_{21}	x_{22}	x_{23}	x_{24}	x_{31}	x_{32}	x_{33}	x_{34}
min z	0	0	-16	4	3	0	0	0	9	-9	-9	0

Because the transportation model seeks to *minimize* cost, the entering variable is the one having the *most positive* coefficient in the z-row. Thus, x_{31} is the entering variable.

(z-row).
 $u + v - c_{ij}$: contribution values

TABLE 5.22 Iteration 1 Calculations

	$v_1 = 10$	$v_2 = 2$	$v_3 = 4$	$v_4 = 15$	Supply
$u_1 = 0$	10 5 - θ	2 10 + θ	20 -16	11 4	15
$u_2 = 5$	12 3	7 5 - θ	9 15	20 5 + θ	25
$u_3 = 3$	4 9	14 -9	16 -9	18 10 - θ	10
Demand	5	15	15	15	



	$v_1 = 10$	$v_2 = 4$	$v_3 = 15$	$v_4 = 15$	Supply
$u_1 = 0$	10 5 - θ	2 10 + θ	20 -16	11 4	15
$u_2 = 5$	12 3	7 5 - θ	9 15	20 5 + θ	25
$u_3 = 3$	4 9	14 -9	16 -9	18 10 - θ	10
Demand	5	15	15	15	

$$\theta \geq 0$$

$$x_{11} = 5 - \theta \geq 0$$

$$x_{22} = 5 - \theta \geq 0$$

$$x_{34} = 10 - \theta \geq 0$$

$$x_{12} = 10 + \theta \geq 0$$

$$5 + \theta \geq 0$$

$$x_{12} = 10 + \theta \geq 0$$

$$\theta \leq 5$$

$$s + \theta \geq 0$$

$$\theta \leq 10$$

$$z = 520 - 5.9$$

$$= 475$$

$$\theta^* = 5 \Rightarrow x_{31} = 5$$

$$v_1 = 1 \quad v_2 = 2 \quad v_3 = 4 \quad v_4 = 15$$

LV: x_{11}

$$u_1 = 0$$

$$u_2 = 5$$

$$u_3 = 3$$

	1	2	3	4	Supply
1	10	2	20	E.V. 11	15
2	12	7	9	20	25
3	5	4	14	16	10
Demand	5	15	15	15	

$$3 + 4 - 1 = 6$$

basic.
(non-empty)

$$u + v \pm c \text{ (Basic)}$$

$$u + v - c : \text{contrib. } z\text{-value}$$

min $z \rightarrow z \text{ value} \rightarrow (+)$
the most.

$$\theta = \min \{15, 10\} = 10$$

$$x_{14} = 10 \text{ (E.V.)}$$

$$x_{24} \rightarrow \text{(L.V.)}$$

$$v_1 = -3 \quad v_2 = 2 \quad v_3 = 4 \quad v_4 = 11$$

$$u_1 = 0$$

$$u_2 = 5$$

$$u_3 = 7$$

	1	2	3	4	Supply
1	10	5	20	10	15
2	12	10	15	4	25
3	5	4	14	5	10
Demand	5	15	15	15	

$$z = 475 - 4 \cdot 10$$

$$z = 435$$

$$= 2 \times 5 + 10 \times 11 + 10 \times 7 + 15 \times 9$$

$$5 \times 4 + 5 \times 18 = 435$$

$$E.V. = ?$$

No E.V. is detected.
So, optimal.

$$x_{12}^* = 5, \quad x_{14} = 10, \quad x_{22} = 10, \quad x_{23} = 15, \quad x_{31} = 5$$

$$x_{34} = 5$$

optimal table

$$x_{34} = 5$$

Optimal table
2-row value = 0
(Nonbasic.

TABLE 5.27 Data for Problem 2

\$5	\$1	\$7	10
\$6	\$4	\$6	80
\$3	\$2	\$5	15
75	20	50	

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1) Balanced?

$$10 + 80 -$$

$10 + 80 - 15 = 65 \neq 75$ → unbalanced.

Row? Column?

5	1	7	10
6	4	6	80
3	2	5	15
0	0	0	40
75	20	50	145

Balanced.

1) N.W. 3

$$v_1 = 5, v_2 = 3, v_3 = 6$$

$$u_1 = 0$$

$$u_2 = 1$$

$$u_3 = -1$$

$$u_4 = -6$$

5	1	7	10
6	4	6	80
3	2	5	15
0	0	0	40
75	20	50	145

$$\frac{4 + 3 - 1}{6} = \frac{6}{6} = 1$$

basic.

$$2 = 560.$$

$$\begin{array}{r|l} 75 & 145 \\ \hline 65 & 145 \end{array}$$

$$\theta = 10 \quad X_{12} = EV = 10$$

$$V_1 = 3 \quad V_2 = 1 \quad V_3 = 4$$

$u_1 = 0$
 $u_2 = 3$
 $u_3 = 1$
 $u_4 = -4$

5	10	7	10
6	4	6	80
3	5	10	15
0	40	0	40
75	20	50	145

$$Z = 560 - 20 = 540$$

$$X_{31}: EV$$

$$\theta = 5$$

$$X_{32}: LV$$

$$V_1 = 3 \quad V_2 = 1 \quad V_3 = 5$$

$u_1 = 0$
 $u_2 = 3$
 $u_3 = 0$
 $u_4 = -5$

5	10	7	10
6	4	6	80
3	2	10	15
0	40	0	40
75	20	50	145

$$Z = 560 - 5 = 555$$

$$= 535 //$$

$$X_{23}: EV$$

$$\theta = 10$$

$$LV: X_{33}$$

$$V_1 = 3 \quad V_2 = 1 \quad V_3 = 3$$

$u_1 = 0$
 $u_2 = 3$
 $u_3 = 0$
 $u_4 = -3$

5	10	7	10
6	4	6	80
3	2	10	15
0	40	0	40
75	20	50	145

alternative

$$Z = 535 - 20 = 515 //$$

$$X_{41} = EV$$

$$\theta = 40$$

$$X_{43} = LV$$

$$\begin{array}{ccc|c} 75 & 20 & 50 & 145 \\ \hline & & & 145 \end{array}$$

alternative opt. sol.

$$X_{43} = LV.$$

Since all 2-row values are nonposi-
tive, we reached the opt.

alternative opt.

5	10	1	7	10
20	6	10	4	30
15	3		2	5
40	0		0	0
75	20	50		145

$z = 515$

Least Cost.

5	10	1	7	10
70	6	4	10	6
5	3	10	2	5
	0		40	0
75	20	50		145

$z = 10 + 420 + 60 + 15 + 20 = 525$

VAM.

Penalty

5	10	1	7	4
				2

$z = 10 +$

5	1	7	100	1
60 ⁶	10 ⁴	10 ⁶	80 ⁷⁰	2
15 ³	2	5	150	1
0	0	40 ⁰	40 ⁰	0

$$2 = 10 +$$

360

40

60

45

515

Optimal.

Penalty:

3

1

5

2
3

1
2

1
1